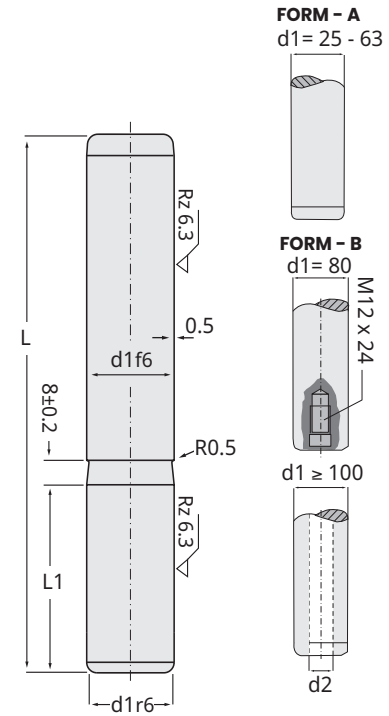
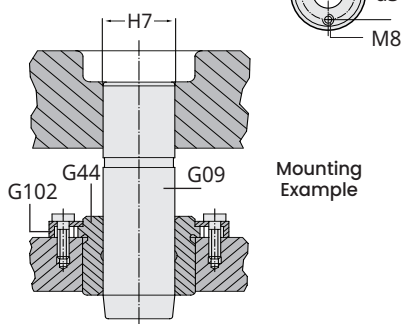


Code: **G09**

Guide Pillar for Large Dies
DIN 9833



DIN 9833 / ISO 9182-3



Mounting Example

Heavy Duty Type for Dies

Practises: Ground surface is holeless up to Ø d = 80 and Ø 80 = M12 X 24 mm has one lifting hole. From Ø d100, it has hollow (d2) and have 2 pieces M8 X 24 mm lifting holes.
Note: Hole tolerance should be H7.
It is recommended to use guide pillar only with self-lubricating guide bushes.
Compatible Bushes:
Self-lubricating bronze guide bushes.

d1	L	d2 Hole	d3	L1	Form
25	125	-	-	40	A
	140				
	160				
	180				
	200				
	224				
32	140	-	-	45	A
	160				
	180				
	200				
	224				
	250				
40	140	-	-	56	A
	160				
	180				
	200				
	224				
	250				
	280				
50	160	-	-	70	A
	180				
	200				
	224				
	250				
	280				
	315				
	355				
63	180	-	-	80	A
	200				
	224				
	250				
	280				
	315				
	355				
	400				

d1	L	d2 Hole	d3	L1	Form
80	200	-	-	100	B
	224				
	250				
	280				
	315				
	355				
	400				
	450				
	500				
100	224	50	72	125	C
	250				
	280				
	315				
	355				
	400				
	450				
	500				
	560				
125	315	65	90	140	C
	355				
	400				
	450				
	500				
	560				
160	400	95	132	180	C
	450				
	500				
	560				

Operating Components:
Self-lubricating guide bushes are recommended.

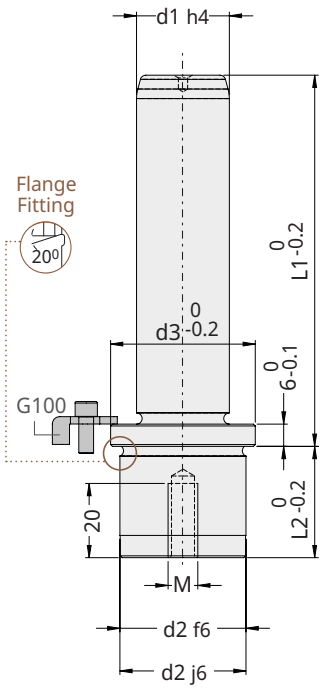
Order: **G09.** d1 x L

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC



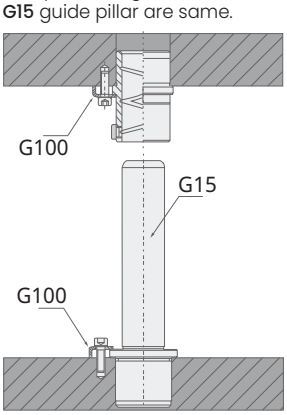
Code: **G15**

Guide Pillar, Thickbacked



Mounting Example

External diameter of bush and rear positioning diameter of the **G15** guide pillar are same.



d1	L1	d2	d3	L2	d4 Clamp	M
20	100	28	34	23	46.2	M8
	125					
	140					
	160					
	180					
	200					
25	100	34	39	30	51.4	M8
	125					
	140					
	160					
	180					
	200					
	220					
	240					
30	125	39	44	37	56.6	M8
	140					
	160					
	180					
	200					
	220					
	240					
	260					
	315					

Flange Fitting: It is perpendicular angle locating on to die plate surface.

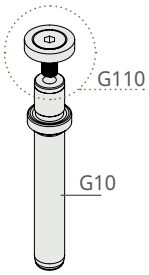
Operating Components: G100
G33- G34 - G35 - G36 - G40 - G42

Order: **G15.** d1 x L1

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Advantageous Product:

It is used as guiding component at dies and blow moulds.
Pillar mounting diameter and bush mounting diameter have the same tolerance value. In machining of guide pillar / bush holes on dies with the advantage of using same cutting tool (drill / reamer) for both guide pillar & bushes hole.
Thus, the necessity of using different cutting tools will disappear.



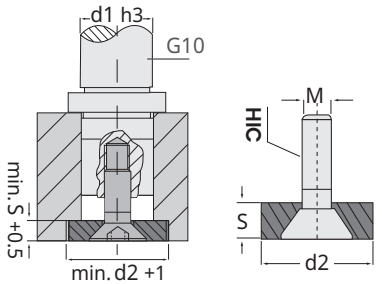


Mounting Flange

Code: **G110**

Guide pillar fixation & Support washer

- Reliable product used for guide pillar with centre collar mountings min. d3 + 1
- It can also be used to support for ejector plates in injection moulds / etc.



d2	d1 Pillar Diameter	S Flange	M Screw
22	Ø 15 / 16	5.5	M8 x 20
25	Ø 19 / 20		
32	Ø 24 / 25		
40	Ø 32 / 30	7.5	
50	Ø 38 / 40	9.5	
60	Ø 48 / 50		
73	Ø 60 / 63		
93	Ø 80	12	M12 x 20

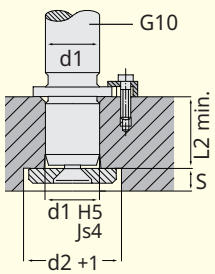
Order: **G110.** d2

Technical Info

Mounting Dimensions and Tolerances of Demountable Die Guide Pillars

Dimensions Given: Recommended (suggestion) for **G10** Guide pillar.

Mounting System: Soft tight fitting system has been selected and determined for **G10** Guide pillar.



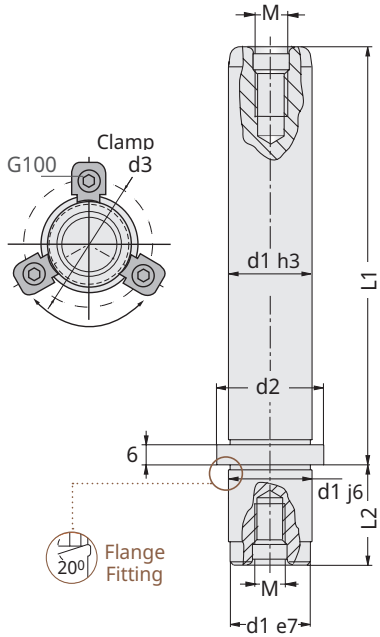
d1	d1 H5	d2 +1	L2	S
15 / 16	+0.008	22	14.5	5.5
19 / 20	+0.009	25	17.5	5.5
24 / 25	+0.009	32	24.5	5.5
30 / 32	+0.011	40	29.5	7.5
38 / 40	+0.011	50	27.5	9.5
48 / 50	+0.013	60	37.5	9.5
60 / 63	+0.013	80	37.5	9.5
80	+0.013	95	48	12



Code: **G10**

Guide Pillar with Centre Collar

DIN 9825 / ISO 9182-5



Flange Fitting: It is perpendicular angle locating on to die plate surface.

Technical Info

It can be inserted directly to the holes which opened on the die plate. During mounting please avoid tight fitting. It can be positioned via soft-tight fitting to the holes and clamps or mounting flange with suitable tolerances or also with chemicals.

* Refer to the mounting measurement tolerance table on the ***Page 63*** for correct fitting of guide pillar to the die plate.

Please do not use two products without oil grooved together. One of the products should be with oil grooved.

Die Producers: Should pay attention to choose products suitable to the tolerances / standards in guide pillar & bush selection.
To prevent incorrect closing of dies, three pieces main dimension and one piece auxiliary dimension can be used:
(15 - 19 - 24 - 32 - 38 - 48 - 60)

For precision and safe mounting: G110 mounting flange.

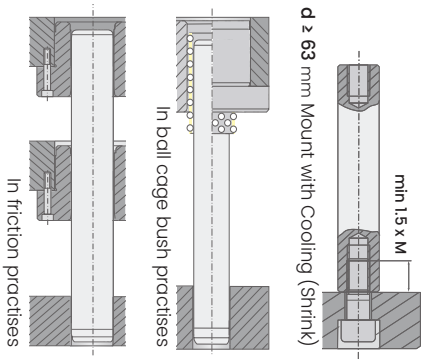
d1	L1	L2	d2	d3 Clamp	M
15 16	100 112 125 140 160 180 200 220 250 300	20	22	33	M8
19 20	100 112 125 140 160 180 200 220 250 300 320	23	25	36.4	M8
24 25	100 112 125 140 160 180 200 224 250 320 350	30	32	44	M8
32 30	112 125 140 160 180 200 224 250 280 315 350 400	37	40	52.4	M8
38 40	125 140 160 180 200 224 250 280 315 355	37	50	62.8	M8
48 50	140 160 180 200 224 250 280 315 355 400	47	63	76	M8
60 63	160 180 200 224 250 280 315 355 400	47	80	93.2	M8
80	200 224 250 280 315 355 400	60	95	108.4	M 12

Operating Components:
All sliding systems also can be used with ball cage bush.

Order: **G10.** d1 x L1

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC



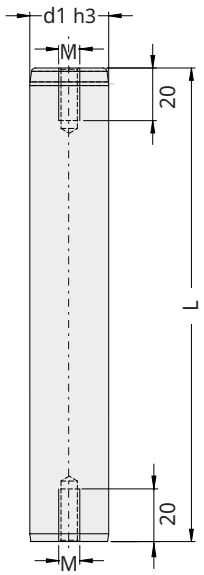


Guide Pillar, Plain & Threaded

Code: **G24**

DIN 9825 / ISO 9182

d1	L	M
NEW 10 12	90 100 112 125 140	-
15 16	90 100 112 125 140 160 180 200	M8
19 20	100 112 125 140 160 180 200 224 250	M8
24 25	100 112 125 140 160 180 200 224 250 280 315	M8
30 32	112 125 140 160 180 200 224 250 280 315	M8
38 40	125 140 160 180 200 224 250 280 315 355 400	M8
48 50	140 160 180 200 224 250 280 315 355 400	M 12
60 63	200 224 250 280 315 355 400	M 12
80	250 280 315 355 400 450 500	1



Technical Info

Section (d), two different products are to avoid incorrect closing of die during mounting, three pieces main dimension and one piece auxiliary should be used:
(d= 15 - 19 - 24 - 32 - 38 - 48 - 60 mm).

If extreme lateral forces are occurred at dies, in these cases self-lubricating wear plates should be used with pillars.

Order: **G24.** d1 x L

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Operating Components:
All sliding systems also can be used with ball cage bush.

3D CAD - Formats: CATIA, STEP, IGES 3D, X_T

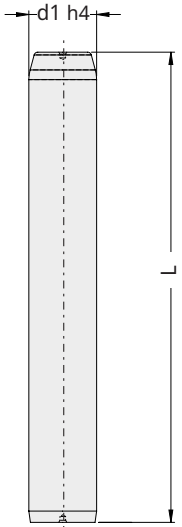


Guide Pillar

Code: **G19**

DIN 9825 / ISO 9182

d1	L
12	100 125
18	125 160
30	160 240

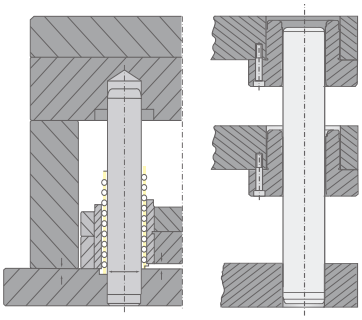


* In-die auxiliary guide pillar

* Guide pillars for ejector plate in injection moulds

Order: **G19.** d1 x L

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC



Technical Info

It is used as auxiliary guiding component in dies or progressive die plates. It can be used with all sliding system or ball cage bush tools. When precision and iterative high speeds are required in injection mould ejector plates, it is suitable to use together with ball cage tools and also with sliding and self-lubricating guide bush components.

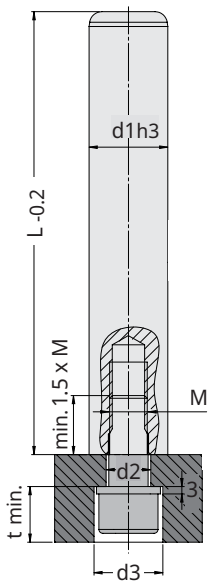


Code: **G166**

Guide Pillar - Bolted Type

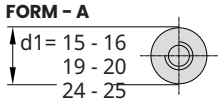
DIN 9825 / ISO 9182 - 2

G166 is supplied with a screw and washer.

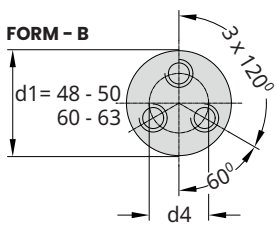


Hole pattern for pillar fastening:

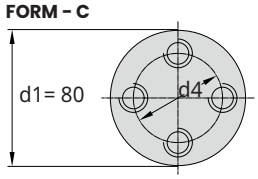
FORM - A



FORM - B



FORM - C



3D CAD - Formats: CATIA, STEP, IGES 3D, X_T

d1	d2	d3	d4	t min.	M	L	Screw	Form
15 16	9	17	-	12	M8	90 100 112 125 140 160 180 200	M8 x 35	A
19 20	11	20	-	14	M10	100 112 125 140 160 180 200 224 250 280 315	M10 x 40	A
24 25	14	22	-	16	M12	100 112 125 140 160 180 200 224 250 280 315	M12 x 40	A
30 32	18	28	-	20.5	M16	125 140 160 180 200 224 250 280 315 355	M16 x 40	A
38 40	18	28	-	20.5	M16	160 180 200 224 250 280 315 355 400 450	M16 x 40	A
48 50	14	22	28	16	M12	180 200 224 250 280 315 355 400 450 500	M12 x 50	B
60 63	18	28	34	20.5	M16	250 280 315 355 400 450 500	M16 x 60	B
80	18	28	54	20.5	M16	280 315 355 400 450 500 550 600	M16 x 60	C

Technical Info

Section (d), two different products are to avoid incorrect closing of die during mounting, three pieces main dimension and one piece auxiliary should be used:
(d= 15 - 19 - 24 - 32 - 38 - 48 - 60 mm).

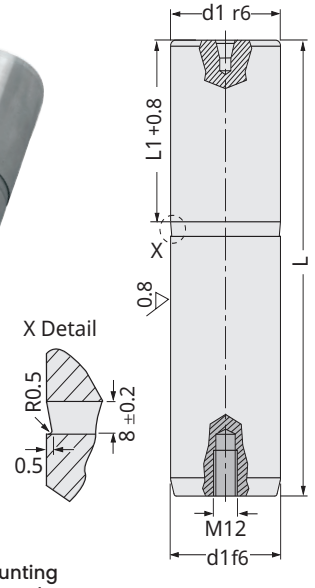
If extreme lateral forces are occurred at dies, in these cases self-lubricating wear plates should be used with pillars.

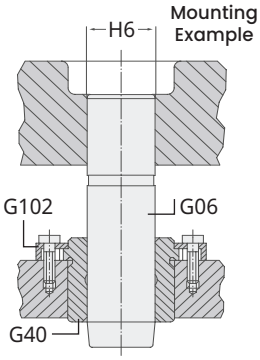
Order: **G166.** d1 x L

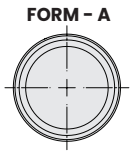
Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

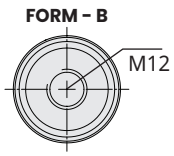
Operating Components:
All sliding systems also can be used with ball cage bush.











Guide Pillar / DIN 9833

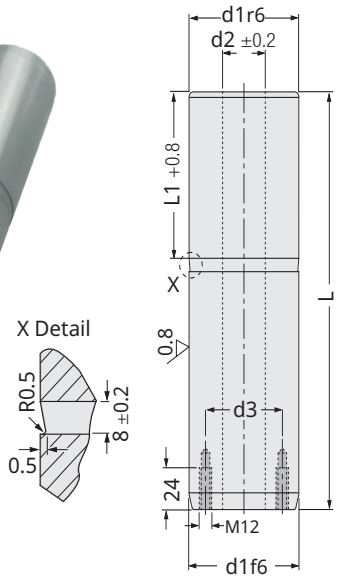
Code: **G06**

d1	L	L1	Form
32	160	45	A
	180		
	200		
40	200	56	A
	224		
	240		
50	224	70	A
	250		
	280		
63	250	80	A
	280		
	315		
80	224	100	A
	250		
	280		
	315		
	355		
100	315	125	A
	355		
	400		
125	355	140	A
	400		
	450		
	500		
	550		

Order: **G06.** d1 x L

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC





Guide Pillar / DIN 9833

Code: **G07**

d1	L	d2	d3	L1
80	250	40	58	100
	280			
	315			
	355			
100	315	50	72	125
	355			
	400			
125	355	65	90	140
	400			
	450			
	500			
	550			

Order: **G07.** d1 x L

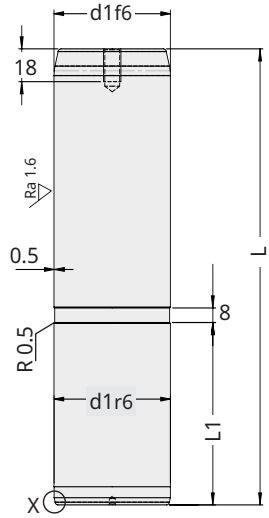
Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC

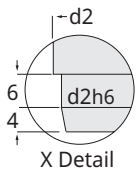
Customised Guide Units

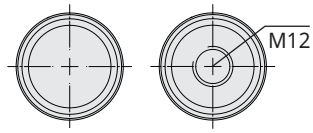
Upon request, TECINMAK manufactures highly complex and precise guide elements based on customer specifications and provides measurement protocols if needed. This includes both the reworking of catalogue products and the production of entirely new items.

3D CAD - Formats: CATIA, STEP, IGES 3D, X_T









Guide Pillar Reference: WDX

Code: **G02**

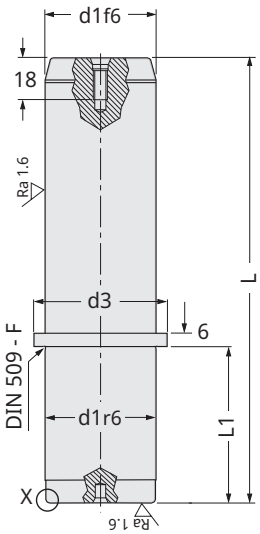
d1	L	L1	Form
25	140	40	A
	160		
	180		
32	140	45	A
	160		
	180		
	200		
40	160	56	A
	180		
	200		
	224		
	250		
50	160	70	A
	180		
	200		
	224		
	250		
	280		
	315		

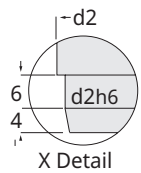
d1	L	L1	Form
63	220	80	A
	224		
	250		
	280		
	315		
	355		
	400		
80	250	100	B
	280		
	315		
	355		
	400		
	500		
100	315	125	B
	355		
	400		
	500		

Order: **G02.** d1 x L

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC







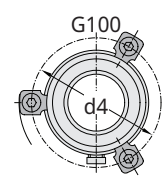
Guide Pillar with Centre Collar

Code: **G01**

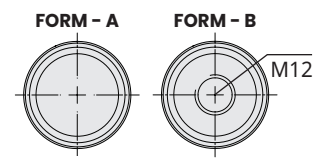
Reference: WDX

d1	L	d3	L1	Form
25	140	33	40	A
	160			
	180			
32	140	40	45	A
	160			
	180			
	200			
40	160	50	56	A
	180			
	200			
	224			
	250			
50	160	60	70	A
	180			
	200			
	224			
	250			
	280			
	315			

d1	L	d3	L1	Form
63	220	80	80	A
	224			
	250			
	280			
	315			
	355			
	400			
80	250	90	100	B
	280			
	315			
	355			
	400			
	500			
100	315	110	125	B
	355			
	400			
	500			



Clamping Detail

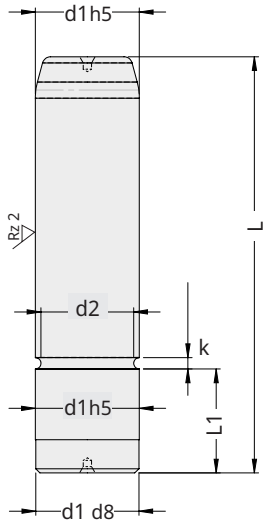


d1	25	32	40	50	63	80	100
d4	45.2	52.4	62.8	73	93.2	103.4	123.4

Order: **G01.** d1 x L

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC





Guide Pillar Reference: CNOMO Code: G03

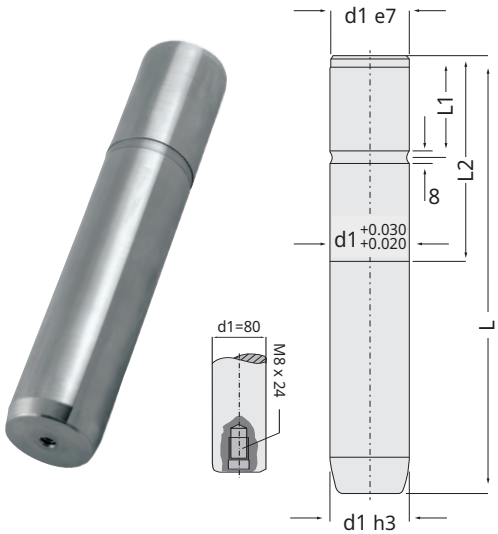
Order Code	d1	L	d2	L1	k
G03 25100	25	100	22.3	25	2.7
G03 25125		125			
G03 25140		140			
G03 25160		160			
G03 25180		180			
G03 25200		200			
G03 25224		224			
G03 32125	32	125	27.8	32	4.2
G03 32140		140			
G03 32160		160			
G03 32180		180			
G03 32200		200			
G03 32224		224			
G03 32250		250			
G03 40180	40	180	35.8	63	4.2
G03 40200		200			
G03 40224		224			
G03 40250		250			
G03 40280		280			
G03 40315		315			

Material: 1.1213 (Cf53) / **Hardness:** 58 - 62 HRC

Order Code	d1	L	d2	L1	k
G03 50220	50	220	45.8	80	4.2
G03 50224		224			
G03 50250		250			
G03 50280		280			
G03 50315		315			
G03 50355		355			
G03 50400		400			
G03 63250	63	250	56.8	100	6.2
G03 63280		280			
G03 63315		315			
G03 63355		355			
G03 63400		400			
G03 63450		450			
G03 63500		500			
G03 80315	80	315	73.8	125	6.2
G03 80355		355			
G03 80400		400			
G03 80450		450			
G03 80500		500			
G03 100355	100	355	93.8	160	6.2
G03 100400		400			
G03 100450		450			
G03 100500		500			



For the compatibility chart of Guide Pillars and Bushes, please refer to the "Guiding Components – Technical Information" section at the beginning of Category 2.



Code: **G20**

Guide Pillar for Cast Block Bush

G20 Guide pillar type using with **G80** rectangular flange cast bush. **In progressive form dies, it is reduced disassembly / assembly period of guide pillar & bush. No need to opening of guide pillar & bush holes in die is required.** It reduces labor cost and also saves time by shortening production period of die.
These are die components consisting of guide bush set inserted into lower and upper case.

d1	L	L1	L2
40	180	37	77
	220		
	260		
	300		
	355		
	400		
50	200	47	95
	240		
	280		
	315		
	355		
	400		
	450		

d1	L	L1	L2
63	240	60	120
	280		
	315		
	355		
	400		
	450		
80	240	60	120
	280		
	315		
	355		
	400		
	450		
	500		

Order: **G20.** d1 x L
Cast Bush: **G79.** d1

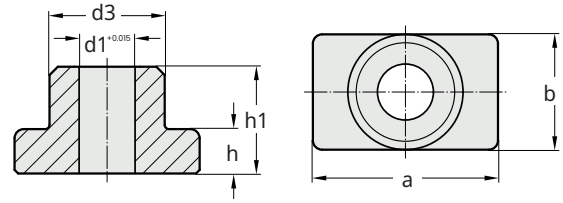
Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC

IMPORTANT:
Check perpendicularity of guide pillars after mounting.



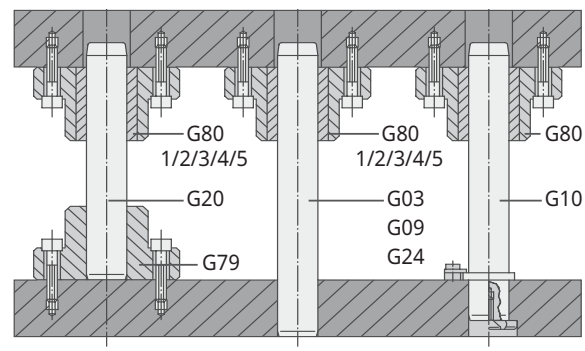
Code: **G79**

Flange Cast Bush - Rectangular



d1	d3	a	b	h	h1
40	80	130	80	30	77
50	96	160	96	35	95
63	110	180	110	35	120
80	130	215	130	40	120

Note: Other guide pillars conforming to rectangular flange cast bush connection.
It can be used with **G03 - G09 - G24 - G10 - G20** in dies.





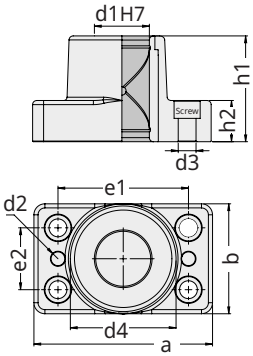
Flange Cast Bush Rectangular
Inner has been honed to Ød1 +0.015
Dimension details of rectangular flange cast bush are same as other bushes (G80..)

Order: **G79.** d1

d1	a	b	h1	h2	d2	d3	d4	e1	e2	Screw
40	130	80	77	30	10	13	70	95	45	12
50	160	96	95	35	10	15	83	118	55	14
63	180	110	120	35	12	17	94	132	62	16
80	215	130	120	40	14	21	112	160	75	20

Order: **G80-1.** d1

Set Upon request
G20 + G79 + G80.1



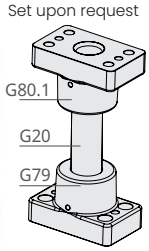
Flange Cast Bush Rectangular

Code: **G80 - 1**



Steel Type

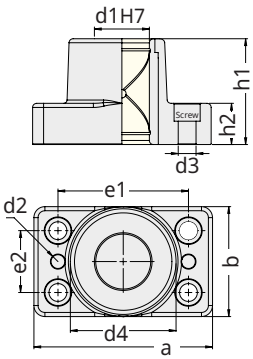
Set upon request



d1	a	b	h1	h2	d2	d3	d4	e1	e2	Screw
40	130	80	77	30	10	13	70	95	45	12
50	160	96	95	35	10	15	83	118	55	14
63	180	110	120	35	12	17	94	132	62	16
80	215	130	120	40	14	21	112	160	75	20

Order: **G80-2.** d1

Set Upon request
G20 + G79 + G80.2



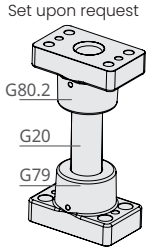
Flange Cast Bush Rectangular

Code: **G80 - 2**



Bronze Type

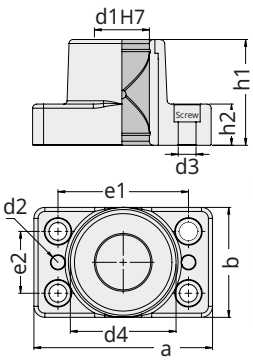
Set upon request



d1	a	b	h1	h2	d2	d3	d4	e1	e2	Screw
40	130	80	77	30	10	13	70	95	45	12
50	160	96	95	35	10	15	83	118	55	14
63	180	110	120	35	12	17	94	132	62	16
80	215	130	120	40	14	21	112	160	75	20

Order: **G80-3.** d1

Set Upon request
G20 + G79 + G80.3



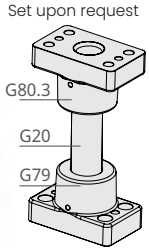
Flange Cast Bush Rectangular

Code: **G80 - 3**



Cast Type

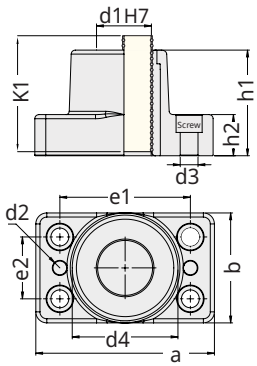
Set upon request



d1	a	b	h1	h2	d2	d3	d4	e1	e2	K1	Screw
40	130	80	77	30	10	13	70	95	45	80	12
50	160	96	95	35	10	15	83	118	55	105	14
63	180	110	120	35	12	17	94	132	62	140	16
80	215	130	120	40	14	21	112	160	75	140	20

Order: **G80-4.** d1

Set Upon request
G20 + G79 + G80.4 + G70



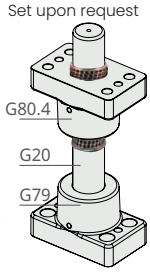
Flange Cast Bush Rectangular

Code: **G80 - 4**



Ball Cage Type

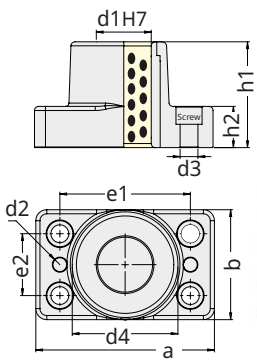
Set upon request



d1	a	b	h1	h2	d2	d3	d4	e1	e2	Screw
40	130	80	77	30	10	13	70	95	45	12
50	160	96	95	35	10	15	83	118	55	14
63	180	110	120	35	12	17	94	132	62	16
80	215	130	120	40	14	21	112	160	75	20

Order: **G80-5.** d1

Set Upon request
G20 + G79 + G80.5



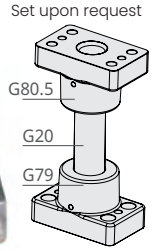
Flange Cast Bush Rectangular

Code: **G80 - 5**



Self-Lubricating Type

Set upon request

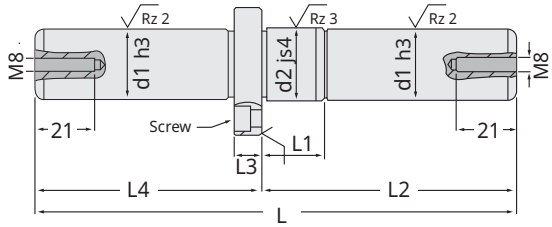


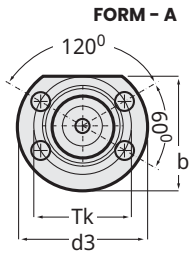


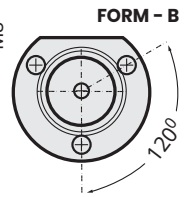
Code: **G135**

Guide Pillar with Centre Flange

(Screw clamping guide pillar with centre collar)







d1	L	L1	L2	L4	L3	d2	d3	b	Tk	Screw	Form
20	150	22	80	70	10	22	44	39	33	M5 4 pcs.	A
	160		90	70							
	180		90	90							
	200		90	110							
25	170	22	90	80	10	26	50	45	38	M5 3 pcs.	B
	190		110	80							
	210		120	90							
	190		90	100							
	200		100	100							
	260		160	100							
	220		100	120							
30	190	27	100	90	10	32	58	53	46	M5 3 pcs.	B
	200		110	90							
	260		160	100							
	220		110	110							
	230		120	110							
	250		120	130							
	260		130	130							
32	190	27	100	90	10	34	60	55	48	M5 3 pcs.	B
	200		110	90							
	260		160	100							
	220		110	110							
	230		120	110							
	250		120	130							
	260		130	130							
40	210	27	110	100	12	42	70	65	56	M6 3 pcs.	B
	220		120	100							
	240		120	120							
	250		130	120							
	270		130	140							
50	250	36	120	130	15	52	80	75	66	M6 3 pcs.	B
	260		110	150							
	270		120	150							

Order: **G135.** d1 x L2 x L

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Guide Pillar with Centre Flange

It can be used in standard pillar systems with many bearing bush connections. No clamping is necessary for die mounting position. It is fixed with hexagon socket head cap screws. The feature of centre flange: The protruding parts near the plate is prevented and a partial empty area is obtained in die mounting. It can be directly mounted in slots drilled on die plate. Shrink fit punchdown should not be done during mounting. It can be inserted with compatible tolerances with medium / shrink fit (bushes) and positioned with screws.

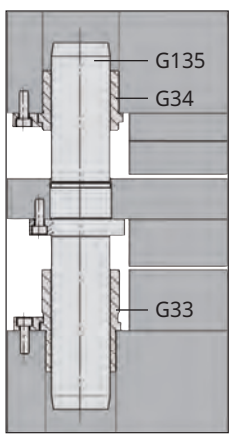
Compatible Bushes

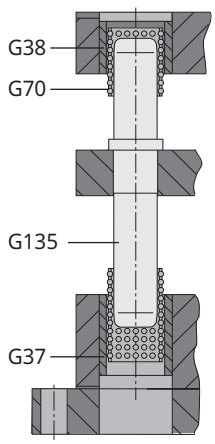


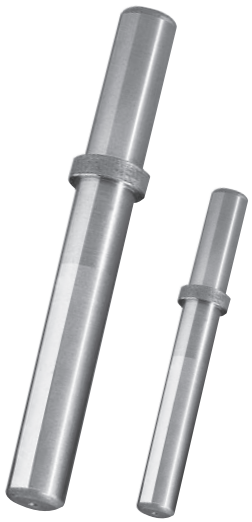


Guide Pillar with Centre Flange
(Ball cage bearing unit)

Dies which are designed with this ball cage bush have high speed and rigid connection.
Hole tolerance for shrink fit is N5.
Fixing place of die guide pillar affects securing lateral load resistance of die bushings.
In dies with guide plates or having guide pillar mounted to die from bottom or top, if the distance (L) applying power is equal, bending and rotary values of guide pillars are equal.
By securing die guide pillar to the guide plate there will be important improvements at bending values of pillars. Until the distance (L / 2) between application point of power and fastening surface reduces to half, load lifting capacity is increased 8 times.

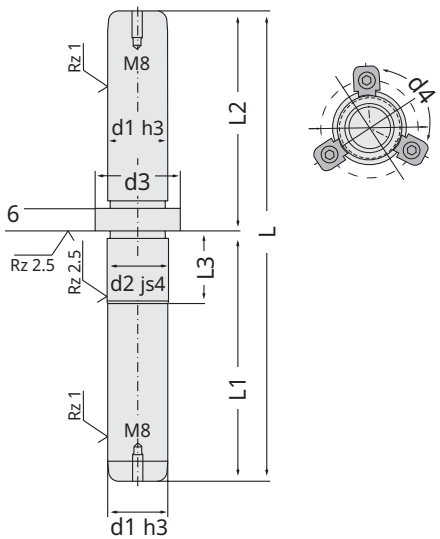






Code: **G08**

Guide Pillar with Centre Fixing



Compatible Bushes



d1	L	L1	L2	L3	d2	d3	d4
20	160	90	70	19	21	26	37.6
	170	100					
	180		80				
	190	110					
	200		90				
	210		100				
25	180	100	80	22	26	32	44
	190	110					
	200		90				
	210	120					
	220		100				
	230		110				
30	180	100	80	25	31	38	50.4
	190	110					
	200		90				
	210	120					
	220		100				
	230	130					
	240		110				
	250	140					
40	200	110	90	27	41	50	62.8
	210	120					
	220		100				
	230	130					
	240		110				
	250	140					

Order: **G08.** d1 x L

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Hole tolerance for shrink fit is N5. Fixing place of die guide pillar affects securing lateral load resistance of die bushings. In dies with guide plates or having guide pillar mounted to die from bottom or top, if the distance (L) applying power is equal, bending and rotary values of guide pillars are equal. By securing die guide pillar to the guide plate, there will be important improvements at bending values of pillars. Until the distance (L / 2) between application point of power and fastening surface reduces to half load lifting capacity is increased 8 times.

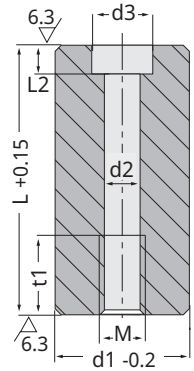


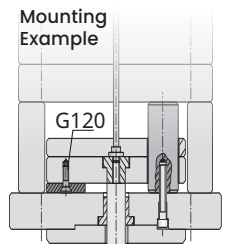
Support Pillar for Injection Mould

(Ejector plate / thrust wedge)

Code: **G21**

d1	L	d2	d3	L2	t1	M
32	47	6.5	11	7.5	15	M8
	57					
	67					
	77					
	87					
	97					
40	47	8.5	15	10	20	M10
	57					
	67					
	77					
	87					
	97					
	117					
50	47	8.5	15	10	20	M10
	57					
	67					
	77					
	87					
	97					
	117					
63	57	8.5	15	10	20	M10
	67					
	77					
	87					
	97					
	117					
	137					
	157					
80	57	10.5	18	12	25	M12
	67					
	77					
	87					
	97					
	117					
	137					
	157					



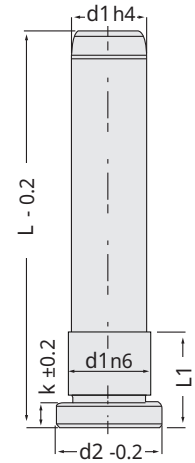


In injection moulds;

The thrust wedge that can be used in order to avoid dent load between support plate (H4) and bottom joint plate (H5A) also can be provided working of ejector plates more rigidly and sensitively. In order to avoid clicking (gap) **G110 mounting flange** can be used.

Order: **G21.** d1 x L





Guide Pillar for Ball Cage

Code: **G18**

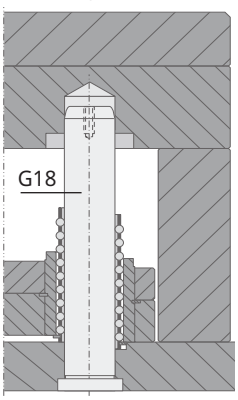
It is used in ejector plates of injection moulds and ejector plates with ball cage bush working precision and serially. In addition, it is compatible to work with intermediate plate of progressive dies / dies as auxiliary centering component.

d1	L	L1	d2	k
NEW 10	80	12	13	4
	100			
	120			
12	80	17	16	4
	100			
	120			
18	120	22	22	6
	140			
	160			
20	120	22	24	6
	140			
	160			
	200			
25	140	27	28	6
	160			
	200			
	240			
30	160	36	36	6
	200			
	240			
40	180	45	48	10
	200			
	240			
	300			

Compatible Bushes



Mounting Example



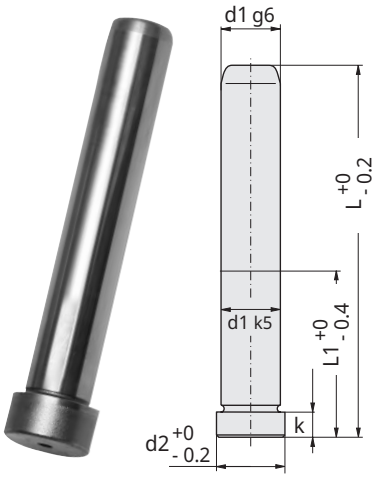
Order: **G18.** d1 x L

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

G18 - Areas of Usage:

* In ejector plates, ball cage bush precision ejector systems.
* It is used in dies as intermediate centering components.

CAD - Formats:
ATIA, STEP, IGES 3D, X_T



Guide Pillar

(without oil groove)

Code: **G17**

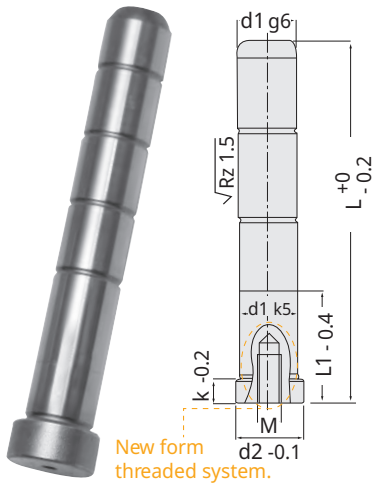
d1	L	L1	d2	k
10	40	13	12	3
	60			
	80			
	100	15		
	120			
12	40	18	16	6
	60			
	80			
	100			
	120	22		
	140			
	160			
14	60	22	18	8
	80			
	100			
	120			
	140	26		
	160			
	180			
16	40	16	20	8
	60	24		
	80			
	100			
	120			
	140	26		
	160			
	180			
	200			
18	60	26	22	8
	80			
	100			
	120			
	140	28		
	160			
	180			
	200			
	240			
20	60	28	24	8
	80			
	100			
	120			
	140			
	160	32		
	180			
	200			
	240			
22	80	37	26	15
	100			
	120			
	140	43		
	160			
	180			
	200			
	220			
	240			
	300			
24	80	39	28	15
	100			
	120			
	140	45		
	160			
	180			
	200			
	220			
	240			
	300			
25	80	35	28	15
	100			
	120			
	140			
	160	44		
	180			
	200			
	220			
	240			
	300			
30	100	45	36	15
	120			
	140			
	160			
	180	53		
	200			
	220			
	240			
	300			
40	160	55	48	15
	180			
	200			
	220	63		
	240			
	280			
	300			
	320			
	360			
50	160	65	58	15
	200			
	240			
	300			
	360			

Order: **G17.** d1 x L

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Operating Components: with all lubricated type bushes in injection moulds /dies.





Guide Pillar
(with oil groove)

Code: **G11**

It can be fixed by thread in cap direction. By disassembling mould / die without dismantling, it provides process iteration advantage. The corners at oil grooves are chamfered (it reduces friction).

d1	L	L1	d2	k	M
10	40	13	12	3	-
	60				
	80				
	100	15			
	120				
12	40	18	16	6	-
	60				
	80				
	100				
	120	22			
	140				
	160				
14	60	22	18	8	-
	80				
	100				
	120				
	140	26			
	160				
	180				
16	40	16	20	8	6
	60	24			
	80				
	100				
	120				
	140	26			
	160				
	180				
	200				
	220				
	250				
	300				
18	60	26	22	8	8
	80				
	100				
	120				
	140	28			
	160				
	180				
	200				
	240				
20	60	28	24	8	8
	80				
	100				
	120				
	140				
	160	32			
	180				
	200				
	220				
	240				
	250				
	300				
	320				
22	80	37	26	15	8
	100				
	120				
	140	43			
	160				
	180				
	200				
	220				
	240				
	300				

d1	L	L1	d2	k	M
24	80	39	28	15	8
	100				
	120				
	140	45			
	160				
	180				
	200				
	220				
	240				
	300				
25	80	35	29	10	8
	100				
	120				
	140				
	160	44			
	180				
	200				
	220				
	250				
	300				
	320				
	350				
30	100	45	36	15	8
	120				
	140				
	160				
	180	53			
	200				
	220				
	240				
	300				
	360				
	400				
40	160	55	48	15	8
	180				
	200				
	220	63			
	240				
	280				
	300				
	320				
	360				
50	160	65	58	15	8
	200				
	240				
	300	73			
	360				

IMPORTANT!

Injection Mould: Our guide pillars & bushes cover standard mould systems. But, guide pillar & bushes with different tolerances should be selected for aluminum injection moulds (high temperature / pressure). It should be requested from our company (production with special materials).

ATTENTION !

Please do not use two products together that have no oil grooves.
One of these products should be oil grooved.

Order: **G11.** d1 x L

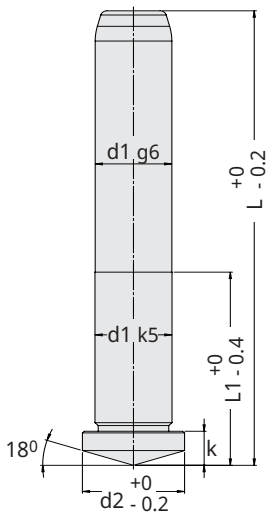
Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Operating Components:
With all unlubricated steel bushes in injection moulds or dies.

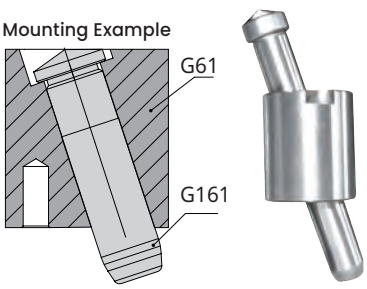


Angled Guide Pillar Code: G161

- **Compatible Guide Bush:** "G61"
- No reworking necessary before mounting.



Mounting Example



Order: **G161.** d1 x L

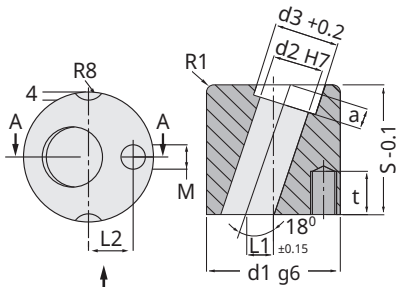
Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

d1	L	L1	d2	k
12	60	34	16	6
	80			
	100			
	120			
	140			
	160			
14	60	34	18	8
	80			
	100			
	120			
	140			
	160			
	180			
16	60	43	20	8
	80			
	100			
	120			
	140			
	160			
	180			
	200			
20	60	43	24	8
	80			
	100			
	120			
	140			
	160			
	180			
	200			
	240			
25	80	52	28	15
	100			
	120			
	140			
	160			
	180			
	200			
	220			
	240			
	300			
30	100	60	36	15
	120			
	140			
	160			
	180			
	200			
	220			
	240			
	300			



Angled Guide Bush Code: G61

(Core pillar holder)



The "**B**" view of **R8** detail.

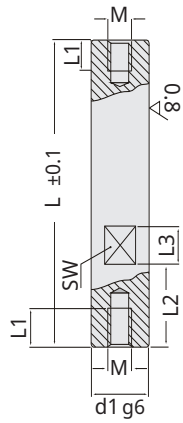
d2	s	d1	a	d3	L1	L2	M	t
12	36	32	6	17	7	10	M6	11
14	36	36	8	19	7	12	M8	16
16	46	40	8	21	8.5	13.5	M8	16
20	46	45	8	25	8.5	15	M8	16
25	56	50	15	29	10	16	M8	16
30	66	63	15	37	12	21	M10	20

Order: **G61.** d2

Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC

This unit eliminates angular perforation requirement. To insert this unit, only a hole is perforated at the front of the mould. **Angled guide bush** can be used with **G161, G14** or **G17** guide pillar.





Ejector Rod

Code: **G16**

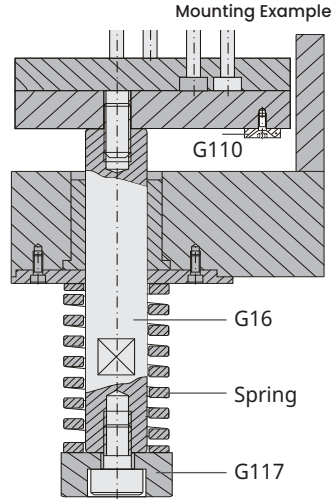
d1	L	M	L1	L2	L3	SW
10	60	M6	10	21	10	8
	70					
	80					
	100					
	140					
14	60	M8	13	21	10	10
	80					
	100					
	120					
	140					
	180					
16	60	M10	16	21	10	12
	70					
	80					
	100					
	120					
	140					
	200					
18	100	M10	20	30	10	14
	120					
	140					
	160					
	180					
	200					
	220					
	240					

d1	L	M	L1	L2	L3	SW
20	120	M12	24	40	16	14
	140					
	160					
	180					
	200					
	240					
24	120	M12	24	40	16	17
	140					
	160					
	180					
	200					
	240					
	260					
34	160	M16	34	50	16	24
	200					
	240					
	260					
	300					

Order: **G16.** d1 x L

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

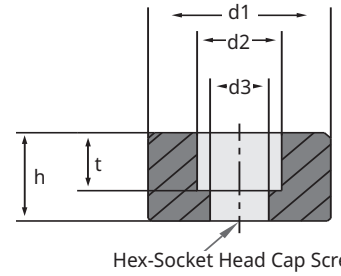
Mounting Example





Code: **G117**

Mounting Washer for Ejector Rod



d1	h	t	d2	d3
15	10	8.4	11	6.6
22	14	11	14.5	9
36	20	13.5	17.5	11
46	25	15.5	20	14
56	32	21	26	18

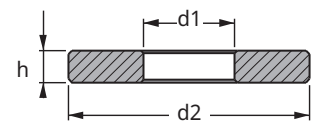
Order: **G117.** d1 x h



Code: **G114**

Thick Mounting Washer

Mould inner mounting washer / thick



Form	d1	d2	h	weight
M6	6.4	17	3	6 gr.
M8	8.4	23	4	10 gr.
M10	10.5	28	4	16 gr.
M12	13	35	5	32 gr.
M14	15	40	5	40 gr.
M16	17	45	6	60 gr.
M18	19	45	6	60 gr.
M20	21	50	6	75 gr.

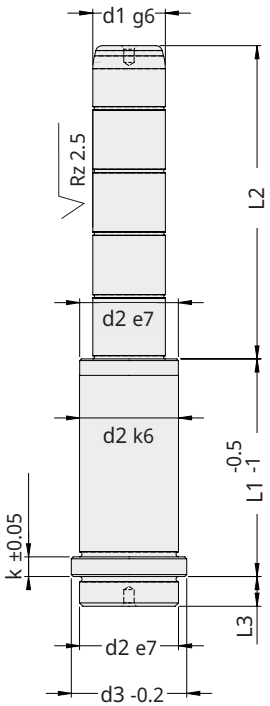
Order: **G114.** Form



Code: **G13D**
(without oil groove)
Upon request

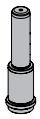
Code: **G13**

Guide Pillar with Centring Collar
(with oil groove & without oil groove)





Code: **G13** (with oil groove)
Guide Pillar with Centring Collar



Code: **G13D** (without oil groove)
Guide Pillar with Centring Collar

d1	L1	L2	L3	d2	d3	k
15 14	17	35 55 75 95	9	20	25	6
15 14	22	30 50 70 90 110 125 150	9	20	25	6
15 14	27	30 45 65 85 105 125 145 165	9	20	25	6
15 14	36	35 55 75 95 115 125 145 155	9	20	25	6
15 14	46	35 45 65 85 105 125 145	9	20	25	6
15 14	56	35 55 75 95 135	9	20	25	6

d1	L1	L2	L3	d2	d3	k
15 14	66	55 65 85 95 105 125	9	20	25	6
15 14	76	35 55 75 95 125	9	20	25	6
15 14	86	55 75 95	9	20	25	6
15 14	96	55 75 95	9	20	25	6
15 14	116	75	9	20	25	6
18 20	17	35 55 75 120	9	26	31	6
18 20	22	35 45 65 85 95 115	9	26	31	6
18 20	27	35 45 65 85 105 125 165 225 245	9	26	31	6

Compatible Bushes



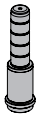
Note:
To avoid wrong closing of mould holder plates, 2 different products are used.
Main dimensions: 14 - 20 - 24 - 30 - 40
other **auxiliary dimensions:**
15 - 18 - 22 - 32 - 42.
To avoid wrong closing of mould holder plates, can be used at 3 main dimensions guide pillar and 1 auxiliary dimension guide pillars. Our mould pillars were polished with surface polishing machine at final stage of production. The products with oil groove are available in our stocks.

Order: **G13.** d1 x L1 x L2
(with oil groove)

Order: **G13D.** d1 x L1 x L2
(without oil groove)

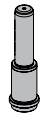
Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Continued on the next page →



Code: **G13** (with oil groove)

Guide Pillar with Centring Collar



Code: **G13D** (without oil groove)

Guide Pillar with Centring Collar

Continued on the next page →

d1	L1	L2	L3	d2	d3	k
18 20	36	35	9	26	31	6
		55				
		75				
		95				
		115				
		135				
		165				
		225				
		255				
18 20	46	35	9	26	31	6
		45				
		65				
		85				
		105				
		135				
		165				
		245				
18 20	56	35	9	26	31	6
		55				
		75				
		95				
		115				
		135				
		155				
18 20	66	35	9	26	31	6
		55				
		75				
		95				
		115				
		135				
		145				
18 20	76	35	9	26	31	6
		55				
		75				
		95				
		115				
		135				

d1	L1	L2	L3	d2	d3	k
18 20	86	55	9	26	31	6
		75				
		95				
		125				
		135				
18 20	96	55	9	26	31	6
		75				
		95				
18 20	116	75	9	26	31	6
		115				
18 20	136	135	9	26	31	6
22 24	17	35	9	30	35	6
		55				
		75				
22 24	22	35	9	30	35	6
		55				
		75				
		95				
		105				
		130				
22 24	27	35	9	30	35	6
		45				
		65				
		85				
		105				
		125				
		165				
		205				
		245				
22 24	36	35	9	30	35	6
		55				
		75				
		95				
		115				

d1	L1	L2	L3	d2	d3	k
22 24	36	135	9	30	35	6
		165				
		205				
		245				
		285				
22 24	46	35	9	30	35	6
		45				
		65				
		85				
		105				
		125				
		165				
		205				
22 24	56	35	9	30	35	6
		55				
		75				
		95				
		115				
		135				
		165				
		205				
22 24	66	35	9	30	35	6
		55				
		75				
		95				
		115				
		135				
		155				
22 24	76	35	9	30	35	6
		55				
		75				
		95				
		115				
		145				
22 24	86	55	9	30	35	6
		75				
		95				
		125				
		135				
22 24	96	55	9	30	35	6
		75				
		95				
		125				

d1	L1	L2	L3	d2	d3	k
22 24	106	55	9	30	35	6
		75				
		95				
		115				
22 24	116	75	9	30	35	6
		115				
		155				
22 24	136	95	9	30	35	6
		135				
32 30	27	35	9	42	47	6
		45				
		65				
		85				
		105				
		125				
		145				
		165				
		185				
		245				
		285				
32 30	36	35	9	42	47	6
		55				
		75				
		95				
		115				
		135				
		155				
		245				
		285				
32 30	46	45	9	42	47	6
		65				
		85				
		105				
		125				
		145				
		165				
		245				
		285				

IMPORTANT !

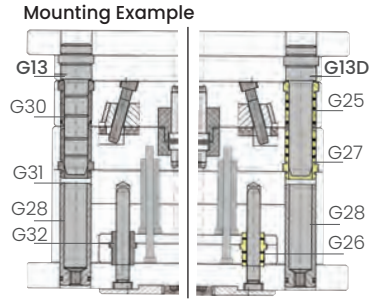
Injection Mould: Our guide pillars & bushes cover standard mould systems. But, guide pillar & bushes with different tolerances should be selected for aluminum injection moulds (high temperature / pressure). It should be requested from our company (production with special materials).

Order: **G13.** d1 x L1 x L2
(with oil groove)

Order: **G13D.** d1 x L1 x L2
(without oil groove)

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC

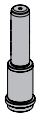
Mounting Example





Code: **G13** (with oil groove)

Guide Pillar with Centring Collar



Code: **G13D** (without oil groove)

Guide Pillar with Centring Collar

d1	L1	L2	L3	d2	d3	k
32 30	56	45 55 75 95 115 135 155 175 245 295	9	42	47	6
32 30	66	45 55 75 95 115 135 155 175 245 295	9	42	47	6
32 30	76	55 75 95 115 135 155 225	9	42	47	6
32 30	86	55 75 95 115 135 155 225	9	42	47	6
32 30	96	55 75 95 115 135 155 205	9	42	47	6
32 30	106	75 95 115 125 145 165	9	42	47	6

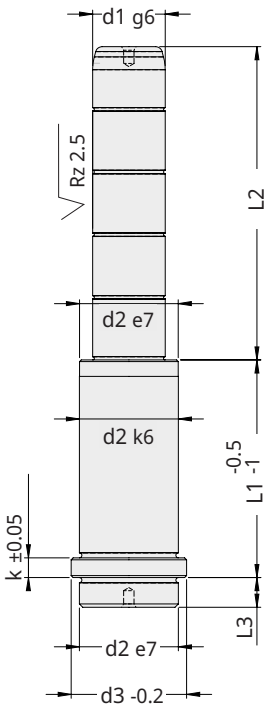
d1	L1	L2	L3	d2	d3	k
32 30	116	75 95 115 135 155	9	42	47	6
32 30	136	95 115 135 155	9	42	47	6
32 30	156	115 155	9	42	47	6
32 30	196	155 195	9	42	47	6
42 40	46	95 165	12	54	60	10
42 40	56	75 115 155 195	12	54	60	10
42 40	66	75 135	12	54	60	10
42 40	76	75 115 175	12	54	60	10
42 40	86	75 135	12	54	60	10
42 40	96	75 115 155	12	54	60	10
42 40	116	95 135 195	12	54	60	10
42 40	136	95 135 215	12	54	60	10
42 40	156	115 155 215	12	54	60	10
42 40	196	155 195 235	12	54	60	10
42 40	246	165 245	12	54	60	10



Code: **G13D**
(without oil groove)
Upon request

Code: **G13**

Guide Pillar with Centring Collar
(with oil groove & without oil groove)



Order: **G13.** d1 x L1 x L2
(with oil groove)

Order: **G13D.** d1 x L1 x L2
(without oil groove)

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Note: To avoid wrong closing of mould holder plates, 2 different products are used.
Main dimensions: 14 - 20 - 24 - 30 - 40 other **auxiliary dimensions:** 15 - 18 - 22 - 32 - 42. To avoid wrong closing of mould holder plates, can be used at 3 main dimensions guide pillar and 1 auxiliary dimension guide pillars. Our mould pillars were polished with surface polishing machine at final stage of production. The products with oil groove are available in our stocks.

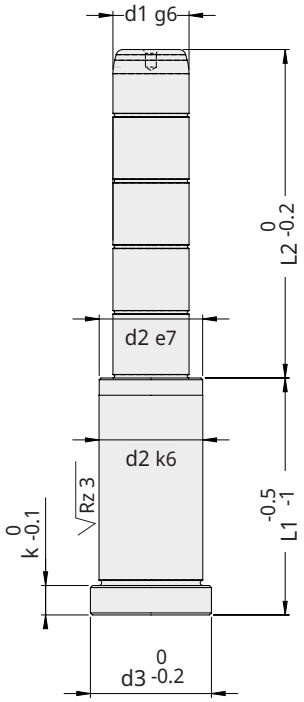
for Turkish Mould Standard



Code: **G12D**
(without oil groove)
Upon request

Code: **G12**

Guide Pillar - Shouldered
(with oil groove & without oil groove)



d1	L1	L2	d2	d3	k
12	26	26 36 46 56	16	19	4
12	36	26 36 46 56	16	19	4
12	46	26 36 46 56 66	16	19	4
14	26	26 36 46 56 66 76	20	25	6
14	36	26 36 46 56 66 86	20	25	6
14	46	26 36 46 56 76 96	20	25	6
14	56	26 36 46 66 86 106	20	25	6
16	26	26 36 46 56 76 96	24	28	6
16	36	26 36 46 56 66 86 106	24	28	6
16	46	26 36 46 56 66 76 96 116	24	28	6
16	56	26 36 46 56 66 76 96 116	24	28	6
16	66	26 36 46 56 66 76 86 106 116	24	28	6
16	76	26 36 46 56 66 76 96 116	24	28	6
16	86	26 36 46 56 66 76 96 116	24	28	6
18	26	26 36 46 56 66 86	26	31	6
18	36	26 36 46 56 66 86 106	26	31	6
18	46	26 36 46 56 66 86 106 126	26	31	6
18	56	26 36 46 56 66 86 106 126	26	31	6
18	66	26 36 56 76 96 116	26	31	6
18	76	26 46 66 86 106 126	26	31	6
18	86	26 46 66 86 106 126	26	31	6
20	26	26 36 46 56 66 86	28	32	6
20	36	26 36 46 56 66 76 96	28	32	6
20	46	26 36 46 56 66 76 86 106	28	32	6
20	56	26 36 46 56 66 76 96 116	28	32	6

Order: **G12.** d1 x L1 x L2
(with oil groove)

Order: **G12D.** d1 x L1 x L2
(without oil groove)

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Continued on the next page ⟶

Compatible Bushes



IMPORTANT !
Injection Mould: Our guide pillars & bushes cover standard mould systems. But, guide pillar & bushes with different tolerances should be selected for aluminum injection moulds (high temperature / pressure). It should be requested from our company (production with special materials).



Code: **G12** (with oil groove)

Guide Pillar - Shouldered



Code: **G12D** (without oil groove)

Guide Pillar - Shouldered

for Turkish Mould Standard

d1	L1	L2	d2	d3	k
20	66	26	28	32	6
		36			
		46			
		56			
		66			
		76			
		96			
		116			
20	76	26	28	32	6
		36			
		46			
		56			
		66			
		76			
		86			
		106			
		126			
20	86	26	28	32	6
		36			
		46			
		56			
		66			
		76			
		86			
		106			
		126			
20	96	26	28	32	6
		36			
		46			
		56			
		66			
		76			
		86			
		106			
		126			
25	26	26	34	38	8
		36			
		46			
		56			
		66			
		76			
		96			
25	36	26	34	38	8
		36			
		46			
		56			
		66			
		76			
		86			
		106			
25	46	26	34	38	8
		36			
		46			
		56			
		66			
		76			
		96			
		116			
25	56	26	34	38	8
		36			
		46			

d1	L1	L2	d2	d3	k
25	56	56	34	38	8
		66			
		76			
		96			
		116			
25	66	26	34	38	8
		36			
		46			
		56			
		66			
		76			
		86			
		106			
		126			
25	76	26	34	38	8
		36			
		46			
		56			
		66			
		76			
		86			
		106			
		126			
25	86	36	34	38	8
		46			
		56			
		66			
		76			
		86			
		96			
		116			
		136			
25	96	36	34	38	8
		46			
		56			
		66			
		76			
		86			
		96			
		116			
		136			
25	106	46	34	38	8
		56			
		66			
		76			
		86			
		96			
		116			
		136			
30	36	36	39	43	10
		46			
		56			
		66			
		86			
		106			
30	46	36	39	43	10
		46			
		56			

d1	L1	L2	d2	d3	k
30	46	66	39	43	10
		76			
		96			
		116			
30	56	36	39	43	10
		46			
		56			
		66			
		76			
		86			
		106			
		126			
30	66	36	39	43	10
		46			
		56			
		66			
		76			
		86			
		96			
		116			
		136			
30	76	36	39	43	10
		46			
		56			
		76			
		96			
		116			
		136			
30	86	36	39	43	10
		46			
		66			
		86			
		106			
		126			
		146			
30	96	46	39	43	10
		66			
		86			
		106			
		126			
		146			
30	106	56	39	43	10
		76			
		96			
		116			
		136			
30	116	56	39	43	10
		76			
		96			
		116			
		136			
40	56	56	54	58	10
		86			
		106			

d1	L1	L2	d2	d3	k
40	56	126	54	58	10
		156			
		186			
40	66	66	54	58	10
		96			
		116			
		136			
		166			
		196			
40	76	76	54	58	10
		106			
		126			
		146			
		166			
		196			
		216			
40	86	86	54	58	10
		126			
		146			
		176			
		206			
		226			
40	96	96	54	58	10
		136			
		156			
		186			
		216			
		236			
40	106	106	54	58	10
		126			
		156			
		186			
		226			
		246			
40	116	116	54	58	10
		136			
		166			
		196			
		236			
		256			
40	126	126	54	58	10
		166			
		196			
		236			
		266			
40	136	136	54	58	10
		196			
		226			
		266			

Order: **G12.** d1 x L1 x L2
(with oil groove)

Order: **G12D.** d1 x L1 x L2
(without oil groove)

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

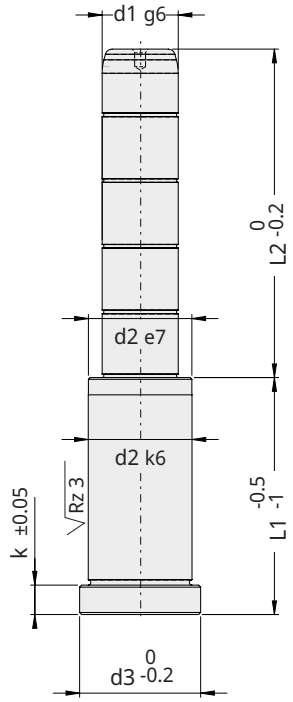


Code: **G50D**
(without oil groove)
Upon request

Code: **G50**

Guide Pillar - Shouldered

(with oil groove & without oil groove)



The tolerances and working space of our products are determined according to **DIN 7161.**

d1	L1	L2	d2	d3	k
14	17	35	20	25	6
		55			
		75			
		95			
14	22	20	20	25	6
		35			
		40			
		45			
		50			
		55			
		65			
		70			
		90			
		110			
14	27	20	20	25	6
		35			
		40			
		45			
		55			
		65			
		85			
		105			
14	36	20	20	25	6
		35			
		40			
		45			
		55			
		65			
		75			
		95			
14	46	20	20	25	6
		35			
		45			
		65			
		85			
		105			
14	56	20	20	25	6
		35			
		55			
		75			
		95			
14	66	55	20	25	6
		65			
		95			
14	76	55	20	25	6
		95			
14	86	55	20	25	6
		95			
14	96	55	20	25	6
		95			
14	116	75	20	25	6

d1	L1	L2	d2	d3	k
15	17	35	20	25	6
		55			
		75			
		95			
15	22	20	20	25	6
		35			
		40			
		45			
		50			
		55			
		65			
		70			
		90			
		110			
15	27	20	20	25	6
		35			
		40			
		45			
		55			
		65			
		85			
		105			
15	36	20	20	25	6
		35			
		40			
		45			
		55			
		65			
		75			
		95			
15	46	20	20	25	6
		35			
		45			
		65			
		85			
		105			
15	56	20	20	25	6
		35			
		55			
		75			
		95			
15	66	55	20	25	6
		65			
		95			
15	76	55	20	25	6
		95			
15	86	55	20	25	6
		95			
15	96	55	20	25	6
		95			
15	116	75	20	25	6

d1	L1	L2	d2	d3	k
18	17	35	26	31	6
		55			
		75			
		95			
18	22	20	26	31	6
		35			
		40			
		45			
		50			
		55			
		60			
		65			
		70			
		80			
		85			
		115			
18	27	20	26	31	6
		35			
		40			
		45			
		50			
		55			
		60			
		65			
		70			
		80			
		85			
		105			
		125			
18	36	20	26	31	6
		35			
		40			
		45			
		50			
		55			
		60			
		65			
		70			
		75			
		80			
		95			
		115			
		135			
18	46	20	26	31	6
		45			
		65			
		85			
		105			
		135			
		165			

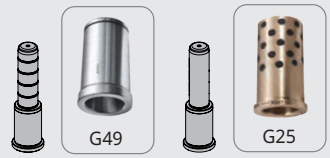
Order: **G50.** d1 x L1 x L2
(with oil groove)

Order: **G50D.** d1 x L1 x L2
(without oil groove)

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Continued on the next page ⟶

Compatible Bushes



IMPORTANT !

Injection Mould: Our guide pillars & bushes cover standard mould systems. But, guide pillar & bushes with different tolerances should be selected for aluminum injection moulds (high temperature / pressure). It should be requested from our company (production with special materials).

d1	L1	L2	d2	d3	k
18	56	20	26	31	6
		35			
		55			
		75			
		95			
18	66	55	26	31	6
		75			
		95			
18	76	55	26	31	6
		75			
		95			
18	86	55	26	31	6
		75			
		95			
18	96	55	26	31	6
		95			
18	116	115	26	31	6
18	136	135	26	31	6
20	17	35	26	31	6
		55			
		75			
		95			
20	22	20	26	31	6
		35			
		40			
		45			
		50			
		55			
		60			
		65			
		70			
		80			
		85			
		115			
20	27	20	26	31	6
		35			
		40			
		45			
		50			
		55			
		60			
		65			
		70			
		80			
		85			
		105			
		125			
20	36	20	26	31	6
		35			
		40			
		45			
		50			
		55			
		60			
		65			
		70			
		75			
		80			
		95			
		115			
		135			

d1	L1	L2	d2	d3	k
20	46	20	26	31	6
		45			
		65			
		85			
		105			
		135			
		165			
20	56	20	26	31	6
		35			
		55			
		75			
		95			
20	66	55	26	31	6
		75			
		95			
20	76	55	26	31	6
		75			
		95			
20	86	55	26	31	6
		75			
		95			
20	96	55	26	31	6
		95			
20	116	115	26	31	6
20	136	135	26	31	6
22	17	35	30	35	6
		55			
		75			
22	22	35	30	35	6
		55			
		75			
		105			
		130			
22	27	25	30	35	6
		45			
		50			
		60			
		65			
		70			
		80			
		85			
		105			
		125			
		165			
22	36	25	30	35	6
		45			
		50			
		55			
		60			
		70			
		75			
		80			
		95			
		115			
		135			
		165			

d1	L1	L2	d2	d3	k
22	46	25	30	35	6
		45			
		50			
		60			
		65			
		70			
		80			
		85			
		105			
		125			
		165			
22	56	25	30	35	6
		45			
		55			
		75			
		95			
		115			
		165			
22	66	55	30	35	6
		75			
		95			
22	76	25	30	35	6
		45			
		55			
		75			
		95			
		115			
22	86	55	30	35	6
		75			
		95			
22	96	55	30	35	6
		75			
		95			
22	116	75	30	35	6
		115			
		155			
22	136	135	30	35	6
22	156	155	30	35	6
24	17	35	30	35	6
		55			
		75			
24	22	35	30	35	6
		55			
		75			
		105			
		130			
24	27	25	30	35	6
		45			
		50			
		60			
		65			
		70			
		80			
		85			
		105			
		125			
		165			

d1	L1	L2	d2	d3	k
24	36	25	30	35	6
		45			
		50			
		55			
		60			
		70			
		75			
		80			
		95			
		115			
		135			
		165			
24	46	25	30	35	6
		45			
		50			
		60			
		65			
		70			
		80			
		85			
		105			
		125			
		165			
24	56	25	30	35	6
		45			
		55			
		75			
		95			
		115			
		165			
24	66	55	30	35	6
		75			
		95			
24	76	25	30	35	6
		45			
		55			
		75			
		95			
		115			
24	86	55	30	35	6
		75			
		95			
24	96	55	30	35	6
		75			
		95			
24	116	75	30	35	6
		115			
		155			
24	136	135	30	35	6
24	156	155	30	35	6

Order: **G50.** d1 x L1 x L2
(with oil groove)

Order: **G50D.** d1 x L1 x L2
(without oil groove)

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

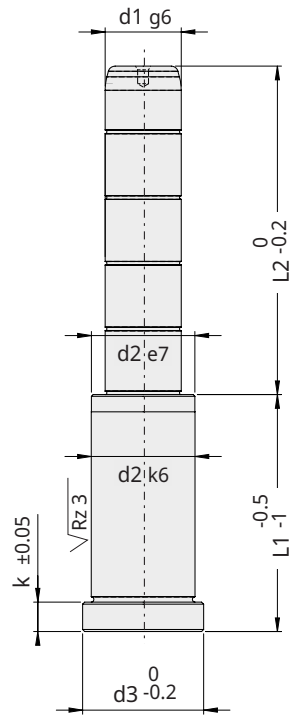


Code: **G50D**
(without oil groove)
Upon request

Code: **G50**

Guide Pillar - Shouldered

(with oil groove & without oil groove)



Order: **G50.** d1 x L1 x L2
(with oil groove)

Order: **G50D.** d1 x L1 x L2
(without oil groove)

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

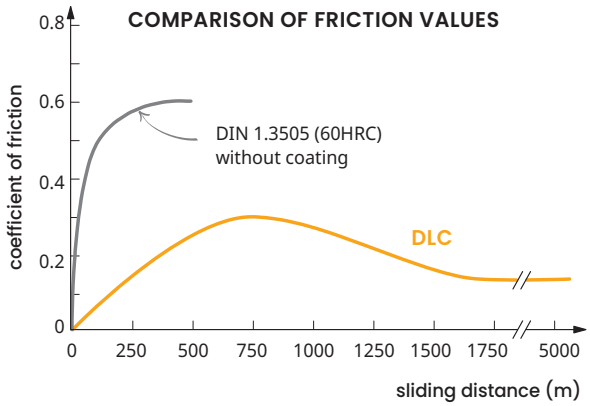
To be continued on the previous page ←

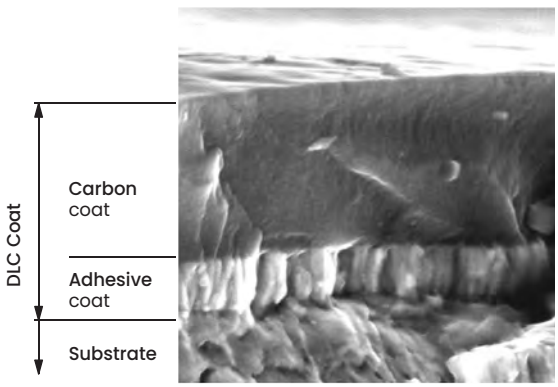
The tolerances and working space of our products are determined according to **DIN 7161.**

d1	L1	L2	d2	d3	k
30	22	35	42	47	6
		75			
		130			
30	27	45	42	47	6
		65			
		105			
		165			
30	36	55	42	47	6
		75			
		95			
		115			
		155			
30	46	45	42	47	6
		65			
		85			
		105			
		125			
		165			
30	56	55	42	47	6
		75			
		95			
		115			
		135			
		175			
30	66	55	42	47	6
		75			
		95			
		115			
		135			
		175			
30	76	55	42	47	6
		75			
		95			
		115			
		155			
30	86	55	42	47	6
		75			
		95			
		115			
		155			
30	96	55	42	47	6
		75			
		95			
		115			
		155			
30	116	75	42	47	6
		115			
		155			
30	136	95	42	47	6
		115			
		155			
30	156	115	42	47	6
		155			
30	176	135	42	47	6
		175			
30	196	155	42	47	6
		195			
32	22	35	42	47	6
		75			
		130			

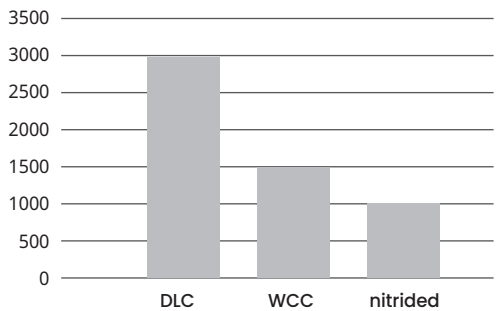
d1	L1	L2	d2	d3	k
32	27	45	42	47	6
		65			
		105			
		165			
32	36	55	42	47	6
		75			
		95			
		115			
		155			
32	46	45	42	47	6
		65			
		85			
		105			
		125			
		165			
32	56	55	42	47	6
		75			
		95			
		115			
		135			
		175			
32	66	55	42	47	6
		75			
		95			
		115			
		135			
		175			
32	76	55	42	47	6
		75			
		95			
		115			
		155			
32	86	55	42	47	6
		75			
		95			
		115			
		155			
32	96	55	42	47	6
		75			
		95			
		115			
		155			
32	116	75	42	47	6
		115			
		155			
32	136	95	42	47	6
		115			
		155			
32	156	115	42	47	6
		155			
32	176	135	42	47	6
		175			
32	196	155	42	47	6
		195			
40	36	75	54	60	10
		135			
40	46	95	54	60	10
		165			
40	56	75	54	60	10
		115			
		155			
		195			
40	66	75	54	60	10
		135			

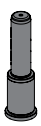
d1	L1	L2	d2	d3	k
40	76	75	54	60	10
		115			
		175			
40	86	75	54	60	10
		135			
40	96	75	54	60	10
		115			
		155			
40	116	95	54	60	10
		135			
		195			
40	136	95	54	60	10
		135			
		215			
40	156	115	54	60	10
		155			
		215			
40	176	135	54	60	10
		155			
		175			
40	196	155	54	60	10
		195			
		235			
42	36	75	54	60	10
		135			
42	46	95	54	60	10
		165			
42	56	75	54	60	10
		115			
		155			
		195			
42	66	75	54	60	10
		135			
42	76	75	54	60	10
		115			
		175			
42	86	75	54	60	10
		135			
42	96	75	54	60	10
		115			
		155			
42	116	95	54	60	10
		135			
		195			
42	136	95	54	60	10
		135			
		215			
42	156	115	54	60	10
		155			
		215			
42	176	135	54	60	10
		155			
		175			
42	196	155	54	60	10
		195			
		235			
50	96	115	66	72	10
	116	135			
	136	135			
	156	155			
	196	175			
60	96	115	80	86	10
	116	135			
	136	135			
	156	155			
	196	175			
	246	195			





Surface hardness HV



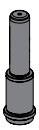


Code: **G50DLC** (DLC Coated)

Guide Pillar - Shouldered

Code: **G13DLC** (DLC Coated)

Guide Pillar with Centring Collar



Customised Guide Units

Upon request, TECINMAR manufactures highly complex and precise guide elements based on customer specifications and provides measurement protocols if needed. This includes both the reworking of catalogue products and the production of entirely new items.

NEW

DLC Coated Guide Pillars



What is DLC ?

DLC means: Diamond like carbon.

The DLC coating is a sliding coating applied using the PACVD (Plasma-Enhanced Chemical Vapor Deposition) process. The coating is deposited through a physical and chemical reaction at approximately 180ºC, with carbon-containing carrier gases introduced simultaneously. Due to the low coating temperature, DLC is also ideal for materials with a low tempering temperature. It combines excellent hardness with superior tribological properties in terms of friction, anti-adhesion, wear, and lubrication.

DLC coating is especially beneficial for applications involving motion or friction in mould making, such as ejection, guiding, and centering. With the DLC coating, it is possible to operate completely lubricant-free..

Advantages:

- Lubricant-free operation
- Extended maintenance intervals
- Very high abrasion resistance
- No seizing, even when running dry
- Good corrosion resistance

Technical Data:

Coating hardness: 2400 - 3000 HV

Coating thickness: ~ 1.5 - 2 µm

Coating temperature: 180ºC

Temperature resistance: 350ºC

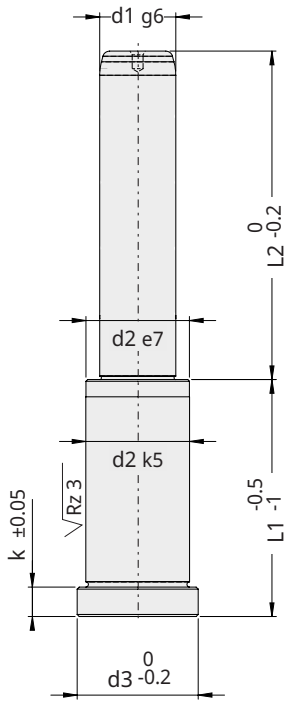
DLC coated high-quality standard components to ensure a longer service life for your moulds.

NEW



Guide Pillar - Shouldered

Code: **G50DLC** (DLC Coated)



Order:
G50DLC. d1 x L1 x L2

Material: 1.7131
Hardness: 60 HRC

Technical Data:

Coating hardness: 2400 - 3000 HV
Coating thickness: ~ 1.5 µm
Coating temperature: 180°C
Temperature resistance: 350°C

DLC coated high-quality standard components to ensure a longer service life for your moulds.

d1	L1	L2	d2	d3	k
14	22	35	20	25	6
		55			
		70			
		90			
14	27	35	20	25	6
		55			
		85			
		105			
14	36	35	20	25	6
		55			
		75			
		95			
14	46	35	20	25	6
		65			
		85			
14	56	55	20	25	6
		75			
14	66	55	20	25	6
		65			
		95			
18	27	35	26	31	6
		55			
		65			
		70			
		80			
		85			
		105			
		125			
18	36	35	26	31	6
		55			
		75			
		95			
		115			
		135			
18	46	45	26	31	6
		65			
		85			
		105			
18	56	55	26	31	6
		75			
		95			
18	66	55	26	31	6
		75			
		95			

d1	L1	L2	d2	d3	k
18	76	55	26	31	6
		75			
		95			
22	36	25	30	35	6
		45			
		55			
		75			
		95			
		115			
		135			
		165			
22	46	25	30	35	6
		45			
		50			
		70			
		85			
		105			
		125			
		165			
22	56	25	30	35	6
		45			
		55			
		75			
		95			
		115			
		165			
22	66	55	30	35	6
		75			
		95			
22	76	55	30	35	6
		75			
		95			
		115			
22	86	55	30	35	6
		75			
		95			
22	96	55	30	35	6
		75			
		95			
30	46	45	42	47	6
		65			
		85			
		105			
		125			
		165			

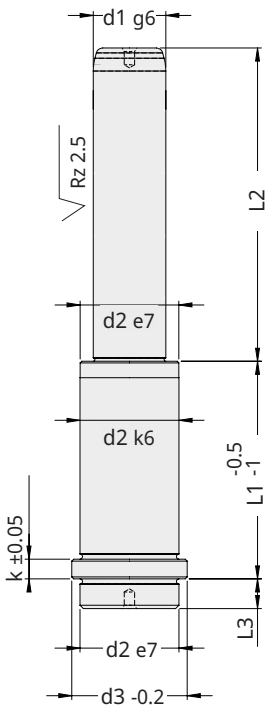
d1	L1	L2	d2	d3	k
30	56	55	42	47	6
		75			
		95			
		115			
		135			
		175			
30	66	55	42	47	6
		75			
		95			
		115			
		135			
		175			
30	76	55	42	47	6
		75			
		95			
		115			
		155			
30	86	55	42	47	6
		75			
		95			
		115			
		155			
30	96	75	42	47	6
		95			
		115			
		155			
30	116	75	42	47	6
		115			
		155			
30	136	95	42	47	6
		115			
		155			
30	156	115	42	47	6
		155			
40	96	75	54	60	10
		115			
		155			
40	116	135	54	60	10
40	136	95	54	60	10
		135			
		215			
40	156	115	54	60	10
		155			
		215			
40	196	155	54	60	10
		195			
		235			

NEW



Guide Pillar with Centring Collar

Code: **G13DLC** (DLC Coated)



Order:
G13DLC. d1 x L1 x L2

Material: 1.7131
Hardness: 60 HRC

Continued on the next page →

Technical Data:

Coating hardness: 2400 - 3000 HV
Coating thickness: ~ 1.5 µm
Coating temperature: 180°C
Temperature resistance: 350°C

d1	L1	L2	L3	d2	d3	k
14 15	22	30 50 70 90	9	20	25	6
14 15	27	30 45 65 85 105	9	20	25	6
14 15	36	35 55 75 95 115	9	20	25	6
14 15	46	35 45 65 85 105 125	9	20	25	6
14 15	56	35 55 75 95 135	9	20	25	6
14 15	66	55 65 85 95 125	9	20	25	6
14 15	76	55 75	9	20	25	6
20 18	22	35 45 65 85	9	26	31	6
20 18	27	35 45 65 85 105 125	9	26	31	6
20 18	36	35 55 75 95 115 135	9	26	31	6
20 18	46	35 45 65 85 105 135	9	26	31	6
20 18	56	35 55 75 95 115	9	26	31	6
20 18	66	35 55 75 95 115	9	26	31	6
20 18	76	55 75 95 115	9	26	31	6
20 18	86	55 75 95 125	9	26	31	6
20 18	96	55 75 95	9	26	31	6
24 22	27	65 85 105 125 165 205	9	30	35	6
24 22	36	35 55 75 95 115 165 205	9	30	35	6
24 22	46	35 45 65 85 105 125 165 205	9	30	35	6
24 22	56	35 55 75 95 115 135 165 205	9	30	35	6
24 22	66	35 55 75 95 115 135 155	9	30	35	6

Coating hardness: 2400 - 3000 HV
Coating thickness: ~ 1.5 µm

Coating temperature: 180°C
Temperature resistance: 350°C



NEW

Code: **G13DLC** (DLC Coated)

Guide Pillar with Centring Collar

d1	L1	L2	L3	d2	d3	k
22	76	55	9	30	35	6
		75				
		95				
		115				
		135				
		155				
22	86	55	9	30	35	6
		75				
		95				
		115				
		135				
		155				
		195				
22	96	55	9	30	35	6
		75				
		95				
		115				
		135				
		155				
		195				
22	116	55	9	30	35	6
		75				
		95				
		115				
		135				
		155				
22	136	75	9	30	35	6
		95				
		115				
30	36	55	9	42	47	6
		75				
		95				
		115				
		155				
		195				
30	46	55	9	42	47	6
		75				
		95				
		115				
		155				
		195				
		235				
30	56	55	9	42	47	6
		75				
		95				
		115				
		155				
		195				
		235				

d1	L1	L2	L3	d2	d3	k
30	66	55	9	42	47	6
		75				
		95				
		115				
		155				
		195				
		235				
30	76	55	9	42	47	6
		75				
		95				
		115				
		155				
		195				
		235				
30	86	55	9	42	47	6
		75				
		95				
		115				
		155				
		195				
		235				
30	96	75	9	42	47	6
		95				
		115				
		155				
		195				
		235				
30	116	75	9	42	47	6
		95				
		115				
		155				
		195				
		235				
30	136	95	9	42	47	6
		115				
		155				
		195				
		235				
30	156	95	9	42	47	6
		115				
		155				
30	176	115	9	42	47	6
		155				
		195				

d1	L1	L2	L3	d2	d3	k
30	196	115	9	42	47	6
		155				
		195				
40	66	75	12	54	59	10
		95				
		115				
		155				
40	76	75	12	54	59	10
		95				
		115				
		155				
		195				
40	86	75	12	54	59	10
		95				
		115				
		155				
40	96	95	12	54	59	10
		115				
		155				
		195				
		235				
40	116	95	12	54	59	10
		115				
		155				
		195				
		235				
40	136	95	12	54	59	10
		115				
		155				
		195				
		235				
40	156	95	12	54	59	10
		115				
		155				
		195				
40	176	115	12	54	59	10
		155				
		195				
40	196	155	12	54	59	10
		195				
		235				

Order:
G13DLC. d1 x L1 x L2

Material: 1.7131
Hardness: 60 HRC

NEW



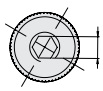
Tolerance
g6

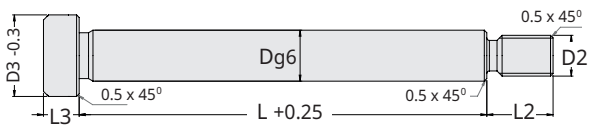
Guiding Shoulder Screw

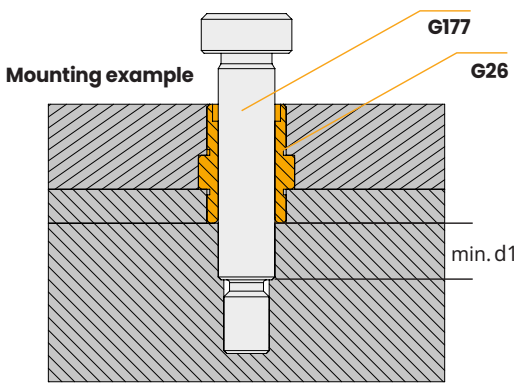
Code: **G177**

* Can be used as a guide bolt with stopper.

* **g6** tolerance









Order Code	D	L	D2	L2	D3	L3	SW
G177.10.25	10	25	M8	13	16	7	5
G177.10.30		30					
G177.10.40		40					
G177.10.50		50					
G177.10.60		60					
G177.10.70		70					
G177.10.80		80					
G177.10.90		90					
G177.10.100		100					
G177.12.25	12	25	M10	16	18	9	6
G177.12.30		30					
G177.12.40		40					
G177.12.50		50					
G177.12.60		60					
G177.12.70		70					
G177.12.80		80					
G177.12.90		90					
G177.12.100		100					
G177.12.120		120					
G177.14.30	14	30	M10	16	21	9	6
G177.14.40		40					
G177.14.50		50					
G177.14.60		60					
G177.14.70		70					
G177.14.80		80					
G177.14.90		90					
G177.14.100		100					
G177.14.120		120					
G177.14.140		140					
G177.18.40	18	40	M12	18	27	11	8
G177.18.50		50					
G177.18.60		60					
G177.18.70		70					
G177.18.80		80					
G177.18.90		90					
G177.18.100		100					
G177.18.120		120					
G177.18.140		140					